



Valve GP- Modulating Test Tool

Maintenance and
Repair

Application

Component Life
Management

Component
Renewal (CRC)

MARC
Management

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November 2021
4.03-211110-1

1.0 Introduction

When transmissions are brought to the workshop to be revised, the Valve GP- Modulating should be tested to see whether it is properly working. Normally, an electrical test is performed. This test measures the current on the solenoid, and if it is between specific values, we believe it works properly. However, this test can only help us decide if the Valve GP- Modulating is working electrically; it does not give a clue about whether or not it is working physically. This means the Valve GP- Modulating could have physical damage, but the electrical test would not be able to detect it, and since we don't test the whole transmission, old valves were assembled without physical testing after revision. That is a huge risk and could lead to other part failures and costly and inefficient rework. Therefore, to detect any kind of physical damage, one of our engineers designed a test tool for the Valve GP- Modulating.

2.0 Best Practice Description

The basic logic behind the test is to simulate the working principle of the Valve GP- Modulating by using this tool. After connecting the Valve GP- Modulating, the pressure gauge and the signal generator to each other, the voltage is applied to the modulation valve in increasing incremental steps. Then the pressure gauge is checked to see whether the pressure is increasing step by step and if it reaches the peak point (about 400 PSI). If it reaches the peak point, it is working properly. If the Valve GP- Modulating couldn't make it to the peak pressure point, that means it is not working properly. You can view the process in the attached video.

3.0 Implementation Steps

1. Create the test tool according to the attached engineering print.
2. Connect the Valve GP- Modulating to the test tool. (Tank and pump lines are specified.)

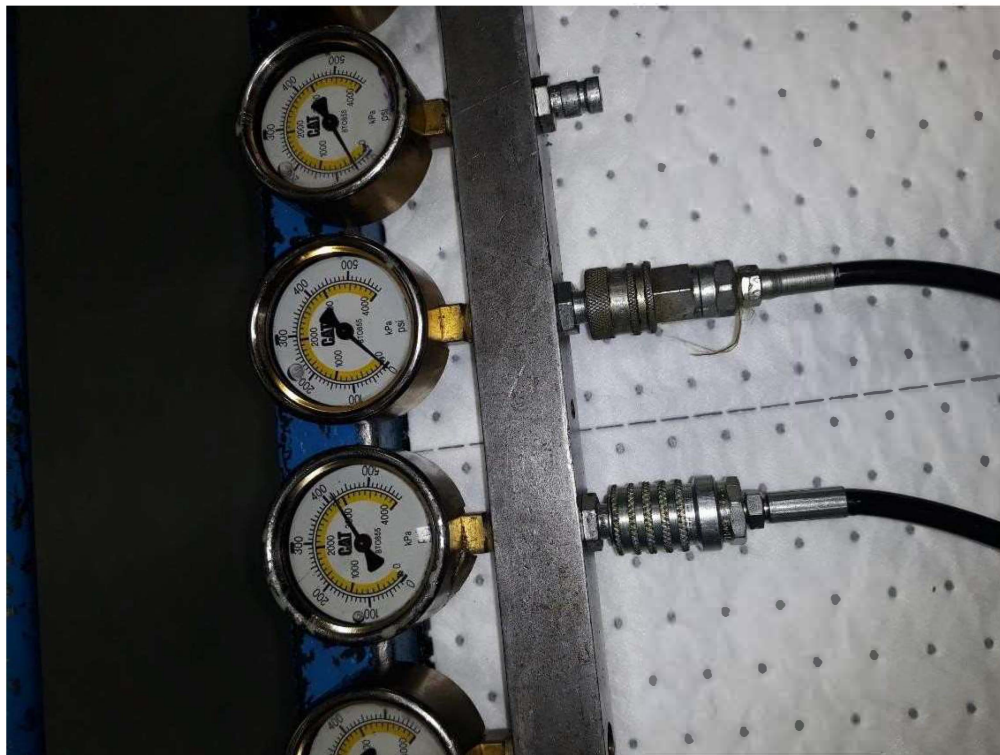


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3. Connect the tank and pump lines.



4. Attach one (1) pressure gauge to the valve to be tested and one (1) pressure gauge to the input line. Attach the signal energy cable.



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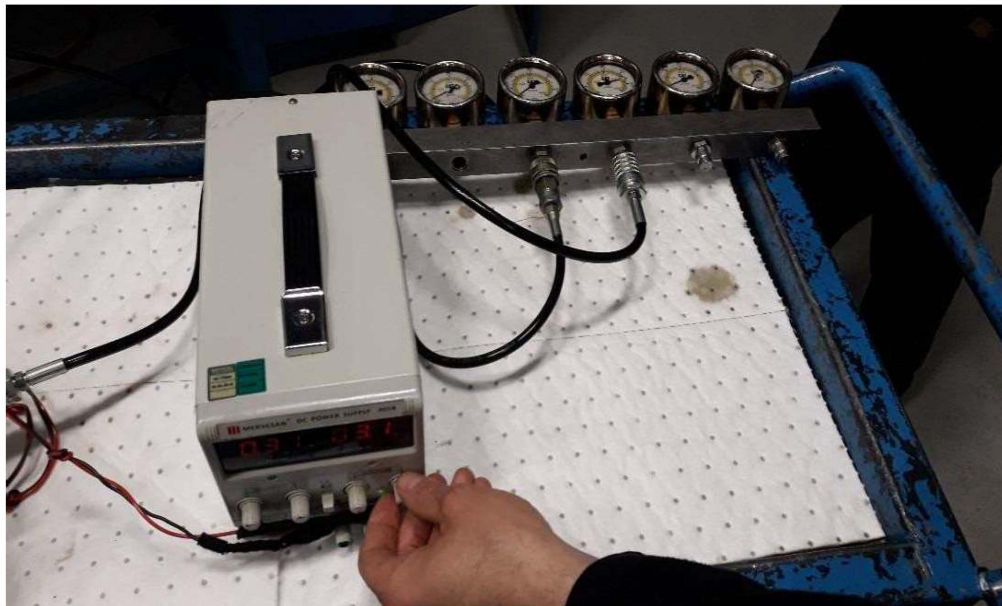
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5. Check the pressure on the modulation by applying voltage (in increasing incremental steps) to the modulation valve and see whether the valve is working or not. The pump inlet pressure should reach 400 PSI.



4.0 Benefits

- Eliminates the risk of reassembling physically damaged valves;
- Prevents damage to other parts of equipment that would be caused by faulty and damaged valves;
- Decreases rework cost and time if the valve is found to be physically damaged;
- Delivers higher quality and more efficient service back to the customer.

5.0 Resources Required

Item	Cost
Aluminum Plate (200x110x80 cm)	\$30
Signal Generator	\$160
Labor	\$60
Total	\$250

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6.0 Supporting Attachments / References

Valve GP- Modulating Test Tool_Technical Drawing.pdf
Valve GP- Modulating Test Tool_Video.mp4

7.0 Related Best Practices

N/A.

8.0 Acknowledgements

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